

Lab program-24:

24. Design a C program to demonstrate UNIX system calls for file management.

Code:

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <fcntl.h>
#include <sys/stat.h>

int main(void) {
    int fd;
    char buffer[100];

    fd = creat("sample.txt", S_IRUSR | S_IWUSR);
    if (fd == -1) {
        perror("creat");
        return 1;
    }
    printf("File 'sample.txt' created successfully.\n");
    close(fd);

    fd = open("sample.txt", O_WRONLY | O_APPEND);
    if (fd == -1) {
        perror("open");
        return 1;
    }
    printf("File 'sample.txt' opened for writing.\n");
```

```
if (write(fd, "Hello, World!\n", 14) == -1) {
    perror("write");
    close(fd);
    return 1;
}
printf("Data written to 'sample.txt'.\n");
close(fd);

fd = open("sample.txt", O_RDONLY);
if (fd == -1) {
    perror("open");
    return 1;
}
printf("File 'sample.txt' opened for reading.\n");

int bytesRead = read(fd, buffer, sizeof(buffer) - 1);
if (bytesRead == -1) {
    perror("read");
    close(fd);
    return 1;
}
buffer[bytesRead] = '\0';
printf("Data read from 'sample.txt':\n%s", buffer);
close(fd);
```

```
if (remove("sample.txt") == -1) {
    perror("remove");
    return 1;
}
printf("File 'sample.txt' deleted.\n");
return 0;
}
```

OUTPUT:

```
File 'sample.txt' created successfully.  
File 'sample.txt' opened for writing.  
Data written to 'sample.txt'.  
File 'sample.txt' opened for reading.  
Data read from 'sample.txt':  
Hello, World!  
File 'sample.txt' deleted.
```

```
-----  
Process exited after 0.02066 seconds with return value 0  
Press any key to continue . . . |
```